Stage-Aware Reinforcement Learning for Vision-Language-Action Manipulation: From LIBERO to Cable-Driven Robots

WORKING MANUSCRIPT — 21 October 2026
Hardware prose assumes successful completion; measurements remain unfilled.

Abstract*—Binary task rewards provide little learning signal when robot rollouts fail at different stages of manipulation. We study stage-aware feedback in two complementary vision-language-action (VLA) reinforcement-learning settings. On LIBERO-Spatial, a phase-aware reward augments terminal verification with bounded credit for new within-phase progress and completed phases. On a cable-driven parallel robot (CDPR), a SmolVLA-conditioned residual controller is bootstrapped from skill experience and optimized with three-stage group-relative credit, protection of achieved milestones from negative completion bonuses, and conditional latent-action likelihoods. In the supplied LIBERO study, dense feedback reaches 65.3% validation success from a 61.0% initialization, compared with 61.7% for terminal-only training on two NVIDIA L40 GPUs. In matched CDPR development evaluation, the 40M continuation achieves 1,258/2,048 strict successes (61.43%), versus 51.56% at 20M; the paired improvement is 9.86 percentage points with a 95% interval of [7.37, 12.30]. These studies examine different implementations and do not constitute a cross-backbone controlled comparison. *[Success-assumption draft: Physical deployment demonstrates complete pick-and-place on the cable robot; trial counts, measured success, and deployment conditions are to be inserted after verification.]

***Index Terms*—Robot learning, reinforcement learning, manipulation, vision-language-action models, cable-driven robots.**

I. Introduction

Pretrained vision-language-action (VLA) models provide a starting point for instruction-conditioned manipulation, but adapting their outputs to a new robot and improving task completion remain separate challenges. A cable-driven parallel robot (CDPR) exposes a particularly concrete adaptation problem: its controller accepts Cartesian translation, yaw, and gripper commands, while a pretrained model inherits the action conventions and experience of other embodiments. Successful placement additionally requires a sequence of behaviors—approach, grasp, lift, transport, and release—whose failures cannot be diagnosed from a terminal outcome alone.

Group-relative reinforcement learning compares sampled trajectories without learning a separate value function. With binary rewards, however, a group containing only failures gives no outcome contrast, even when some trajectories nearly complete the task. Intermediate feedback can expose this progress. It also creates a second problem: improving an intermediate objective need not improve complete-task success, and penalizing an unsuccessful episode can suppress an already competent prefix.

We investigate this tradeoff through two complementary studies. The first adds phase-aware dense shaping to OpenVLA-OFT training in SimpleVLA-RL on LIBERO-Spatial. The second adapts SmolVLA to cable-driven manipulation through skill bootstrapping and staged residual reinforcement learning. Both retain a terminal task verifier, but their intermediate signals differ: dense progress rewards change trajectory returns, whereas staged sparse credit assigns milestone-specific advantages to portions of a trajectory. The CDPR study also addresses likelihood consistency when exploration contains clipped actions and temporally persistent offsets.

Shared question: how can intermediate progress help complete-task RL?

LIBERO / OpenVLA-OFT: phase tracker → bounded dense reward + terminal verifier → LoRA policy updates.

CDPR / SmolVLA: skill experience → residual initialization → protected stage credit + latent likelihood → full-task controller.

Evaluation: benchmark validation; matched CDPR scenes; physical trials.

Fig. 1. The two study branches share a credit-assignment question, while retaining distinct policies, action distributions, and evaluation protocols. Physical image slots are provided in Fig. 4.

Our contributions are a joint account of these two stage-aware feedback designs; a precise CDPR residual-policy and credit formulation with conditional mathematical analysis; and an empirical study combining LIBERO reward comparisons, matched CDPR development evaluation, and a reserved physical deployment study. The current evidence establishes observed improvements within each protocol. It does not establish that either intermediate-feedback design dominates the other, or that the VLA action prior alone explains the CDPR gain.

II. Related Work

VLA adaptation. OpenVLA [1] and SmolVLA [2] provide pretrained visuomotor policies with different architectures and deployment costs. OpenVLA-OFT studies efficient adaptation through action chunking, parallel decoding, and continuous-action supervised objectives [3]. The LIBERO implementation considered here starts from an OpenVLA-OFT checkpoint but applies PPO/GRPO updates to discrete action-token likelihoods. The CDPR branch instead freezes the VLA during the reported continuation and learns a bounded residual controller. These parameterizations require different probability models.

Residual and group-relative learning. Probe, Learn, Distill (PLD) combines residual reinforcement learning with autonomous data generation and distillation [4]. Accordingly, residual learning and success-filtered consolidation are established ingredients, rather than standalone novelty claims here.

Our CDPR emphasis is the interaction between complete-task resets, stage-specific credit, and the executed action process. GRPO originates in language-model reinforcement learning [5]; SimpleVLA-RL brings group-relative online optimization to VLA rollouts [6]. Existing binary-GRPO analyses motivate examining which groups actually provide reward contrast [7], [8].

Intermediate rewards and bounded actions. Potential-based shaping provides conditions for preserving optimal policies [9]. The phase-aware reward below is an engineering shaping rule and is not asserted to satisfy those conditions. Likewise, protecting achieved stages is a credit modification rather than a general policy-improvement guarantee. Work on clipped-action gradients establishes the importance of distinguishing a latent sample from its executed clipped value [10]. We apply that distinction to persistent exploration in the CDPR controller and delimit what it implies for the complete optimization procedure.

III. Problem Formulation

Let c specify a scene and instruction, h_t the observation and controller history, and $Y(\tau) \in \{0,1\}$ the complete-task outcome of trajectory τ. The evaluation objective is

$$J(\theta) = \mathbb{E}_{c,\tau\sim P_\theta(\cdot|c)}[Y(\tau)]. \tag{1}$$

Training may use simulator-derived intermediate predicates, but final evaluation uses the declared task verifier. We distinguish the exploration policy used for training from the deployment policy used for evaluation.

The CDPR action $a_t \in [-1,1]^5$ controls Cartesian translation, yaw, and the gripper. The policy receives camera observations, proprioception, and an instruction; simulator object geometry is used to calculate rewards and evaluation events. Episodes begin with an empty gripper and require the entire manipulation sequence. Strict success requires a physical grasp and held lift, intentional release, and native placement, without latched carry-slip or wrong-placement events. Native placement alone is therefore an insufficient endpoint. The exact event predicates, temporal latches, and abort handling are part of the evaluation contract.

LIBERO success is the environment's binary goal-verification outcome [11]. Its phase tracker uses simulator state during training. This endpoint and the CDPR strict endpoint are reported separately; their percentages cannot be pooled into a common success rate. Similarly, repeated development scenes measure performance on a declared distribution rather than unseen-task generalization.

IV. Stage-Aware Learning

A. CDPR Policy and Bootstrapping

A frozen SmolVLA model produces a mapped prior chunk p. The residual input x combines proprioception and frozen visual features; the residual network also receives p. Its bounded mean is

$$\mu_\theta(x,p) = \tanh(p + s_r \tanh f_\theta(x,p)). \tag{2}$$

Both nonlinearities are retained: s_r bounds the correction before the outer action squashing. The inspected configuration uses $s_r = 1$, eight-slot chunks with replanning every four executed slots, six proprioceptive dimensions, and a 512-dimensional frozen visual projection. The long continuation trains the residual and exploration parameters while keeping inherited VLA/LoRA weights frozen. **[To complete: Confirm these dimensions and controller frequency from the selected run's resolved configuration.]**

Initialization combines skill acquisition, success-filtered residual supervised fitting, and continuous teacher-chain experience before full-task reinforcement learning. This supplies a starting distribution of behaviors, rather than assuming that isolated skill success composes automatically. No newly collected target-robot teleoperation dataset is required by this branch; pretrained robot data and historical scripted/oracle assistance remain part of its lineage. **[To complete: Record exact initializer, experience-bank sizes, filtering criteria, and inherited interaction cost.]**

B. Protected Stage Credit

The full-task collector records three binary milestones: approach, pickup, and strict placement, denoted $M_{i1}, M_{i2}, M_{i3} = Y_i$ for candidate i. Within each scene group, milestone returns are centered and divided by their population standard deviation, with numerical flooring and configured advantage clipping. Let A_{ik} denote this advantage and U_{ik} the corresponding usability mask. Constant-return groups provide no milestone contrast. Invalid candidates are excluded from training statistics, although numerical failures remain failures in evaluation.

For approach and pickup records, a terminal bonus supplements local credit:

$$\widetilde{A}_{ik} = U_{ik}A_{ik} + \beta U_{i3} T_\kappa(A_{i3}, M_{ik}), \tag{3}$$

$$T_\kappa(a,m) = \begin{cases} \kappa a, & m = 1,\ a < 0, \\ a, & \text{otherwise.} \end{cases} \tag{4}$$

Placement uses $U_{i3}A_{i3}$ without an additional duplicate bonus. The inspected settings use $\beta = 1$ and $\kappa = 0$: a failed completion does not contribute a negative terminal bonus to an achieved earlier milestone. This protects one credit component, not every parameter supporting the prefix.

With fixed batch advantages, nonnegative stage/candidate weights w_r, and per-record likelihood ratios ρ_r, the clipped surrogate is

$$S(\theta) = \sum_r w_r \min\{\rho_r \widetilde{A}_r, \mathrm{clip}(\rho_r, 1-\epsilon_-, 1+\epsilon_+)\widetilde{A}_r\}. \tag{5}$$

The weights balance retained stage mass and candidate trajectory lengths. Entropy and mean-action penalties accompany the surrogate, and Adam and gradient clipping determine the actual update. Thus the implemented objective is more specific than the sum of three milestone rewards.

C. Conditional Latent-Action Likelihood

At each sampled slot, exploration produces

$$u_t = \mu_\theta(x_t, p_t) + b_t + \sigma_\theta \odot \varepsilon_t, \qquad a_t = \mathrm{clip}(u_t, -1, 1), \tag{6}$$

where ε_t is standard Gaussian and b_t is a gated contribution from an exogenous episode-level offset. The collector retains the unclipped sample u_t, the realized offset, and the old log probability. New/old ratios score the same sample under the conditional Gaussian density $\mathcal{N}(\mu_\theta + b_t, \operatorname{diag} \sigma_\theta^2)$.

Scoring the clipped action as a Gaussian observation incorrectly represents probability mass at the bounds. Treating a shared offset as fresh independent noise at every step also loses temporal dependence. Conditioning on its recorded realization avoids both mismatches. It does not eliminate biases introduced by stage selection, invalid-candidate filtering, or repeated surrogate optimization.

D. Phase-Aware Dense Feedback on LIBERO

The complementary LIBERO branch represents supported pick-and-place behavior with five ordered phases: reach, grasp, lift, move, and place. Each environment maintains its own tracker. If $p_t \in [0, 1]$ denotes active-phase progress and B_t the best progress previously reached in that phase, the rewarded increment is $\Delta p_t^+ = \max(p_t - B_t, 0)$. The progress baseline resets on phase advancement. Within a phase, revisiting an already reached state earns no additional progress credit.

Let c_t count newly completed phases and y_t denote terminal verifier success. The additive reward is

$$r_t = \lambda y_t + \operatorname{clip}(w_p \Delta p_t^+ + w_c c_t, -C, C). \tag{7}$$

Only shaping is clipped; the original terminal reward is retained without a second terminal term. Unlike CDPR's protected milestone advantages, this rule changes the rewards from which trajectory advantages are constructed. **[To complete: Insert exact phase predicates, progress functions, transition/reset logic, and chunk-to-reward aggregation from source code.]**

Rollout selection is a separate mechanism. A group of n trajectories with terminal outcomes Y_i is retained when $L \le n^{-1} \sum_i Y_i \le U$. Bounds excluding zero reject all-failure groups even if dense returns differ. Dense feedback therefore does not automatically remove this filtering bottleneck. Both the reward law and the acceptance rule must be reported when interpreting update throughput.

V. Mechanism Analysis

A. Latent Scores and Binary Contrast

Assume fixed dynamics, a frozen prior, an exogenous offset law, and regularity allowing differentiation under the expectation. The augmented trajectory density then factors into exogenous terms and conditional latent-policy terms. Consequently,

$$\nabla J = \mathbb{E}\left[Y \sum_t \nabla \log \pi_\theta(u_t \mid h_t, p_t, b_t) \right]. \tag{8}$$

To see this, differentiate the trajectory density: only the policy factors depend on θ at fixed recorded history, while clipping is a deterministic execution map. The offset density cancels in likelihood ratios. A learned covariance contributes its own Gaussian score. Equation (8) describes the full-trajectory score objective, not an unbiasedness guarantee for the stage-filtered PPO surrogate [10].

For G conditionally independent candidates of one scene with binary milestone probability p, the probability of a nonconstant group is

$$P_{\text{mixed}} = 1 - p^G - (1 - p)^G. \tag{9}$$

This follows by excluding the all-success and all-failure outcomes. At $G = 8$, changing p from 0.03 to 0.30 increases this probability from 21.6% to 94.2%. This explains why an intermediate milestone can supply contrast when completion is rare. An almost certain milestone also loses contrast. For heterogeneous scenes, the prediction is the mean of the scene-specific probabilities, rather than the formula evaluated at the overall success rate. These observations are consistent with existing binary-group analyses [7], [8]. Dense within-phase returns can distinguish trajectories sharing a binary outcome, provided the rollout filter retains them.

B. What Prefix Protection Guarantees

Consider a benchmark of three sequential Bernoulli gates with separate logits, $q_k = \operatorname{sigmoid}(\theta_k)$ and $J = q_1 q_2 q_3$. A reached gate produces one decision X_k, with score $X_k - q_k$. Use sign-preserving binary group normalization, nonnegative weights, on-policy raw ascent, $\beta \ge 0$, and $\kappa = 0$, without regularization or optimizer momentum. Then every realized batch direction has $d_k \ge 0$, and

$$J(\theta + \eta d) \ge J(\theta), \qquad \eta \ge 0. \tag{10}$$

Indeed, an achieved gate receives nonnegative milestone credit and no negative completion bonus; a failed gate receives nonpositive credit. In either case credit and score have the same sign. Their weighted product is nonnegative, so every gate logit can only increase. This is a sufficient explanation of the protection rule, not evidence that symmetric credit must fail in this model.

The neural controller shares parameters across stages and assigns many actions to each stage, so the sign argument no longer applies. For an objective with an L-Lipschitz gradient on the actual update segment,

$$J(\theta + \Delta) - J(\theta) \ge \nabla J^\top \Delta - \tfrac{L}{2} \|\Delta\|^2. \tag{11}$$

The inequality follows by integrating the gradient along that segment. Positive alignment and a sufficiently small update would imply improvement, but neither an alignment bound nor L has been measured for the CDPR policy. PPO clipping alone does not establish this condition [12]. The analysis also gives no automatic guarantee for hardware transfer or for deployment with a different exploration law.

VI. Experimental Setup

A. Cable-Driven Manipulation

The simulation study uses a five-channel CDPR in MuJoCo Warp/MJ-Lab, camera inputs, four object categories (apple, orange, potato, tomato), and plate/bowl destinations. The matched evaluation uses 512 distinct development scenes,

excluding the 128-scene in-run validation panel, with four repeats per checkpoint. Each episode has a 128-decision horizon and no settling decisions or teacher/yaw assistance. Deployment uses the residual mean with VLA prior noise scale 1. Numerical failures remain in all-attempt denominators. **[To complete: Supply simulator version, scene-generation ranges, camera calibration, and the resolved action timing.]**

The main comparison evaluates the 20M and 40M latent-likelihood bounded-residual checkpoints, with a panel-selected 39.8M checkpoint as a secondary comparison. These counters measure cumulative selected actions since latent-likelihood conversion; they exclude earlier inherited training and do not count all sampled interactions. The 20M–40M leg adds 19,987,881 selected actions, 609,792 episodes, and 299 updates over 89.14 recorded training hours. The learning rate remains 10^{-5} and the recorded LoRA change is zero.

Reported effect intervals use a percentile bootstrap of per-scene mean differences, keeping repeats clustered by scene. Within-scene permutation p-values use the recorded Holm correction for the primary strict comparisons. Subgroup results are exploratory. This is one continuing training lineage; four evaluation repeats are not four independent training seeds. Final-test results and controlled terminal-only, likelihood, and prior-contribution ablations remain to be added.

B. LIBERO-Spatial Reward Study

The supplied study uses OpenVLA-OFT in SimpleVLA-RL on the ten LIBERO-Spatial tasks, two NVIDIA L40 GPUs, and an aggregate rollout batch of 32. LoRA rank and alpha are both 16, applied to the language-model projector; gradient checkpointing and FSDP offloading reduce memory demand. These updates optimize discrete action-token likelihoods, despite the continuous-action supervised formulation associated with OFT. **[To complete: Confirm implementation commit, tokenizer/action decoding, adapted modules, group size, and optimization schedule.]**

All aggregate reward configurations start from 61.0% validation success. The primary dense condition uses $(w_p, w_c, C, \lambda) = (0.2, 0.3, 0.05, 5)$; its terminal-only control disables shaping and retains $\lambda = 5$. The filtered dense condition additionally uses $[L, U] = [0.1, 0.9]$. With $\lambda = 1$, dense shaping uses $w_p = w_c = 0.05$, while phase-only shaping uses $w_p = 0, w_c = 0.2$; both use $C = 0.05$.

The source reports final recorded evaluations, balanced across the ten tasks. A separate low-success study restricts training and validation to `task_1`, starting at 10%, with $[L, U] = [0.1, 1.0]$ in both conditions. Exact evaluation counts, seeds, task identity, elapsed budgets, and accepted-update counts are absent from the supplied PDF. Until these records are available, its percentages are descriptive results, without inferred confidence intervals or equal-interaction claims.

C. Physical Deployment

[Success-assumption draft: The following protocol and result prose are prepared for a completed successful experiment; retain only details confirmed by the actual trial record.] We evaluated the selected policy on the physical cable-driven platform using **[To complete: hardware configuration]**, **[To complete: camera/state-estimation setup]**, and a control frequency of **[To complete: Hz]**. The deployment condition was **[To complete: direct transfer, calibration only, or hardware adaptation; describe exactly]**. The hardware checkpoint was **[To complete: path and hash]**; it must not be inferred from the simulation table.

Each trial began with an empty gripper and an object outside the destination. A successful trial required acquisition, sustained lift and transport, deliberate release, and placement in the instructed receptacle without assistance. We recorded all attempts, interventions, and aborts under **[To complete: trial/reset protocol and stopping rule]**. The evaluation comprised **[To complete: objects, destinations, layouts, sessions, and trials per policy]**, with **[To complete: baseline and ordering]**. The earlier outline's 80-trial design is only a suggestion, not a completed sample size.

TABLE I
LIBERO-SPATIAL RESULTS TRANSCRIBED FROM THE SUPPLIED DRAFT. ALL ROWS START AT 61.0%; Δ IS IN PERCENTAGE POINTS. EVALUATION COUNTS, SEEDS, AND UNCERTAINTY ARE NOT SUPPLIED.

Configuration	Success (%)	Δ
Primary dense, $\lambda = 5$	65.3	+4.3
Filtered dense $[0.1, 0.9]$	64.7	+3.7
Phase-only, $\lambda = 1$	62.3	+1.3
Terminal-only, $\lambda = 5$	61.7	+0.7
Dense, $\lambda = 1$	61.0	0.0

VII. RESULTS

A. Phase-Aware Feedback on LIBERO

Table I transcribes the supplied study's reward comparison. Primary dense feedback increases validation success from 61.0% to 65.3%, compared with 61.7% for terminal-only reward: a 3.6-percentage-point endpoint difference. The filtered dense configuration reaches 64.7%. The weaker-terminal configurations reach 62.3% and 61.0%, showing that intermediate feedback does not invariably improve the starting policy. Because shaping weights also change, this comparison does not isolate the effect of the terminal coefficient.

In the low-success task study, terminal-only and dense training reach 11% and 18%, respectively, from the same 10% initialization (Table II). The supplied account also reports faster accepted-batch assembly and more optimizer updates for dense feedback during the observed interval. Without timestamps, attempt counts, and a common budget endpoint, this remains a qualitative throughput observation. It does not establish a numerical speedup or improvement per optimizer update.

B. Full-Task CDPR Performance

The 40M endpoint completes 1,258 of 2,048 attempts (61.43%), compared with 1,056 (51.56%) at 20M under the same development protocol (Table III). The gain is 202 successes, or 9.86 percentage points, with a scene-paired 95% interval of [7.37, 12.30] and Holm-adjusted $p = 0.00010$. Counts

TABLE II
REPORTED LOW-SUCCESS LIBERO TASK STUDY. BOTH CONDITIONS USE GROUP-ACCURACY FILTERING $[0.1, 1.0]$. VALUES ARE PERCENTAGES; EVALUATION TIMESTAMPS AND DENOMINATORS REMAIN TO BE SUPPLIED.

Evaluation	Terminal-only	Dense
Initial	10	10
Intermediate	8	12
Final	11	18

TABLE III
MATCHED CDPR DEVELOPMENT EVALUATION: 512 SCENES × FOUR REPEATS PER CHECKPOINT. NF DENOTES NON-FINITE EPISODES. ALL RATES INCLUDE FAILURES IN THE DENOMINATOR.

Checkpoint	Strict count	Strict (%)	Native (%)	NF (%)
20M	1056/2048	51.56	56.74	1.32
40M final	1258/2048	61.43	66.46	1.37
39.8M peak	1256/2048	61.33	66.89	1.76

are compared before rounding. This documents improvement with continued training, rather than an isolated causal effect of the likelihood change.

The panel-selected 39.8M checkpoint achieves 1,256/2,048 (61.33%). Its direct difference from final40m is -0.10 points, with a paired interval of $[-2.05, 1.81]$ and $p = 0.9598$. Thus its higher in-run peak does not provide a demonstrated advantage on the separate development scenes; this is not an equivalence test. Figure 2 shows the unsmoothed in-run trajectory separately from the matched endpoint comparison.

The retained earlier 2M pilot provides additional development context: the bounded-residual continuation achieves 843/2,048 (41.16%) versus 756/2,048 (36.91%) for its source, a paired gain of 4.25 points [2.10, 6.40]. These earlier repeats are not combined with the 20M/40M comparison to infer a single paired learning curve.

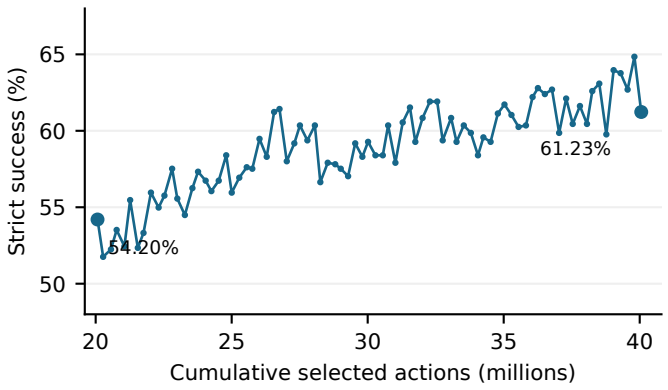


Fig. 2. The 20M–40M continuation on the repeated in-run validation panel (1,024 episodes per reading). These correlated readings describe one training lineage; they are distinct from Table III's 512-scene matched comparison. The horizontal axis counts selected actions since conversion, excluding inherited training.

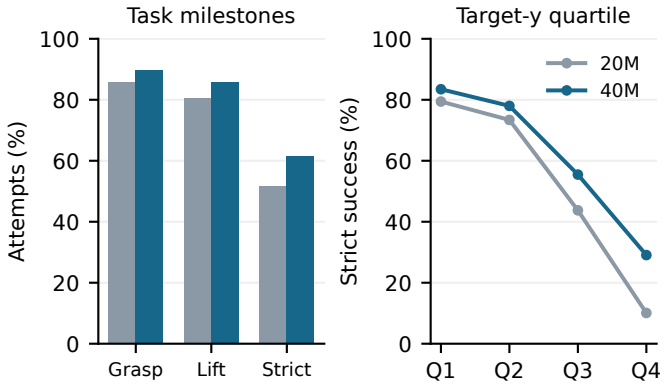


Fig. 3. Matched development diagnostics for 20M and 40M: (left) marginal grasp, held-lift, and strict outcomes over all 2,048 attempts; (right) strict outcome by target-y quartile, with 130, 125, 128, and 129 scenes, each repeated four times. These are descriptive rates; confidence intervals for the main paired effect appear in the text.

C. Stage Retention and Remaining Failures

Continued CDPR training improves physical-grasp and held-lift rates from 85.84% and 80.52% to 89.65% and 85.89%. Strict completion conditional on held lift rises from 64.04% to 71.52% as a descriptive ratio. Carry-slip events decrease from 28.32% to 23.83%, and wrong-placement events from 25.00% to 19.63%. Their paired difference intervals are $[-6.79, -2.25]$ and $[-7.62, -3.17]$ points, respectively. Event flags may overlap and must not be stacked into an exclusive failure partition.

Plate and bowl strict rates at 40M are 68.58% and 54.22%. The far-side target-y quartile improves from 10.08% to 29.07%, but remains substantially weaker than the nearer quartiles (Fig. 3). Potato remains the weakest object category at 46.11%. These exploratory breakdowns locate residual difficulty; they do not establish robustness to new objects or workspace distributions.

Non-finite episodes occur in 1.32% and 1.37% of the 20M and 40M evaluations. The difference interval $[-0.73, 0.83]$ points does not establish equivalence or absence of numerical failures. In the earlier 2M pilot, adding a separate correction network yields 37.55% strict success and 3.56% non-finite episodes, with no demonstrated strict gain over its source: 0.63 points $[-1.76, 3.03]$. This arm is not the architecture producing the 61.43% result.

Matched terminal-only versus staged credit, protected versus symmetric terminal bonuses, legacy versus latent likelihood, and action-prior controls remain outstanding. The existing continuation results cannot substitute for these ablations.

D. Finite-Model Mechanism Checks

Independent finite-model enumeration verifies the binary-group identities for $G = 2, 4, 8$ at seven probabilities with maximum absolute error 1.2×10^{-15}. All 336 enumerated sequential-gate groups at $G = 2, 3, 4$ satisfy the protected-credit sign property; checks over three parameter vectors and three raw-ascent step sizes preserve success. These tests validate the stated finite models, not the neural robot policy.

In a two-decision toy process with persistent offsets, the conditional latent score agrees with the analytic and finite-difference derivatives. At $\theta = 0.1$, the derivative is

TABLE IV
PHYSICAL EVALUATION SLOTS. ALL MEASUREMENTS ARE PENDING; DASHES ARE NOT ZERO OUTCOMES. TRANSFER/ADAPTATION CONDITION AND SAMPLE SIZES MUST ACCOMPANY THE COMPLETED TABLE.

Policy	Successes/trials	Interventions	Time (s)
Selected controller	—	—	—
Matched baseline	—	—	—

(a) Physical CDPR setup
Insert `figures/hardware/setup.jpg`

(b) Recorded rollout: approach, grasp, lift, transport, release
Insert `figures/hardware/sequence.jpg`

Fig. 4. Reserved physical experiment panels. Add actual photographs, frame timestamps, instruction, object/destination, trial ID, and deployment condition. The intended successful sequence is part of the success-assumption draft and remains unverified.

0.3170, whereas treating exact one-step marginals as independent yields 0.6344. Policy-record likelihood audits, measured mixed-group frequencies, and independent update-alignment measurements are still needed to connect these mechanisms to the reported robot improvement.

E. Real-World Manipulation

[Success-assumption draft: This subsection is written under the author's assumption of successful completion. It is not a report of verified hardware outcomes.] The deployed controller completed the instructed pick-and-place sequence on the physical cable-driven robot, progressing from an empty-gripper start through acquisition, lift, transport, and deliberate release. Figure 4 reserves the apparatus and time-ordered rollout panels. The demonstrated behavior establishes feasibility on the tested apparatus; its reliability is quantified by the complete trial inventory in Table IV.

Across **[To complete: N]** attempts, the selected policy achieved **[To complete: S]** strict successes (**[To complete: rate and 95% interval]**). The baseline achieved **[To complete: successes/trials under the same conditions]**. We observed **[To complete: interventions and aborts]**, and successful trials required **[To complete: completion-time summary]**. The dominant residual failure was **[To complete: measured category and count]**; **[To complete: failure sequence/video]** documents its manifestation. These quantities must come from the trial log, including unsuccessful attempts.

The simulation and hardware outcomes use corresponding task-completion criteria, but differ in sensing, contacts, actuation, and reset variability. The transfer description must specify calibration and any hardware adaptation. A successful sequence alone does not establish zero-shot transfer, a particular success percentage, or superiority to the baseline.

VIII. DISCUSSION AND LIMITATIONS

The studies support a common practical motivation: intermediate feedback can distinguish useful partial behavior when terminal verification supplies little contrast. Their evidential roles differ. LIBERO provides a reward comparison within one reported small-compute setup; the CDPR evidence shows continued improvement of a staged residual controller. Different backbones, action likelihoods, tasks, and compute budgets prevent interpreting the two as a controlled comparison of dense and sparse feedback.

Stage awareness also leaves objective mismatch unresolved. A reward can favor repeated partial competence over rare completion, and protecting scalar prefix credit cannot eliminate interference in a shared network. Conditional latent scoring fixes the recorded sampling model but does not certify the full selected, normalized, and clipped optimizer. The finite-model analysis should therefore be interpreted alongside measured learning outcomes, with neural-policy alignment remaining an open diagnostic.

The CDPR results reuse development scenes and one training lineage. Independent seeds, a locked final test, and matched credit/likelihood/prior ablations are needed for stronger attribution and generalization claims. LIBERO lacks trial denominators and quantitative throughput records in the supplied draft. The physical success narrative is a writing assumption until trial evidence is inserted. Historical scripted initialization, manual rewards, calibration, inherited compute, and limited object/destination coverage further constrain claims of autonomous adaptation. Action-prior ablations are especially important because pretrained visual features and pretrained action knowledge can contribute differently.

IX. CONCLUSION

We studied stage-aware feedback for VLA manipulation through phase-aware dense rewards on LIBERO and protected staged residual learning on a cable-driven robot. The supplied LIBERO comparison reports 65.3% validation success for primary dense feedback versus 61.7% for terminal-only training. Matched CDPR development evaluation establishes 61.43% strict success at 40M selected actions, a 9.86-point improvement over 20M. These results motivate intermediate credit while preserving complete-task evaluation as the empirical criterion. *[Success-assumption draft: Successful physical execution extends the feasibility demonstration to the target robot; the final conclusion will state the verified trial-level outcome and deployment condition.]* Controlled ablations and completed hardware records will determine the final scope of the claims.

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